

8.2 Polyphase Implementation of Decimation and Interpolation Filters Using MATLAB

Code:

```
clc;
```

```
clear;
```

```
close all;
```

```
%% Input Signal
```

```
Fs = 1000;          % Sampling Frequency
```

```
t = 0:1/Fs:1;       % Time Vector
```

```
% Test Signal
```

```
x = sin(2*pi*50*t) + 0.5*sin(2*pi*120*t);
```

```
%% FIR Low-Pass Filter Design
```

```
N = 20;             % Filter Order
```

```
Fc = 0.4;           % Normalized Cutoff Frequency
```

```
h = fir1(N, Fc, hamming(N+1));
```

```
%% Decimation Factor
```

```
M = 2;
```

```
% Polyphase Decimation
```

```
y_dec = upfirdn(x, h, 1, M);
```

```
%% Interpolation Factor
```

```
L = 2;
```

```
% Polyphase Interpolation
```

```
y_int = upfirdn(y_dec, h, L, 1);
```

```
%% Adjust Length for Comparison
```

```
if length(y_int) > length(x)
```

```
    y_int = y_int(1:length(x));
```

```
elseif length(y_int) < length(x)
```

```
    x = x(1:length(y_int));
```

```
end
```

```
%% Time Axes
```

```
t1 = (0:length(x)-1)/Fs;
```

```
t2 = (0:length(y_dec)-1)/(Fs/M);
```

```
t3 = (0:length(y_int)-1)/Fs;
```

```
%% BEST GRAPH OUTPUT
```

```
figure('Name','Polyphase Decimation and Interpolation', ...  
      'NumberTitle','off');
```

```
% Original Signal
```

```
subplot(4,1,1)
plot(t1,x,'b','LineWidth',1.5)
title('Original Input Signal')
xlabel('Time (s)')
ylabel('Amplitude')
grid on
```

```
% Decimated Signal
subplot(4,1,2)
stem(t2,y_dec,'r','filled')
title('Decimated Signal ( $M = 2$ )')
xlabel('Time (s)')
ylabel('Amplitude')
grid on
```

```
% Interpolated Signal
subplot(4,1,3)
plot(t3,y_int,'k','LineWidth',1.5)
title('Interpolated/Reconstructed Signal ( $L = 2$ )')
xlabel('Time (s)')
ylabel('Amplitude')
grid on
```

```
% Comparison
subplot(4,1,4)
```

```

plot(t1,x,'b','LineWidth',1.5)
hold on
plot(t3,y_int,'r--','LineWidth',1.5)
title('Comparison of Original and Reconstructed Signals')
xlabel('Time (s)')
ylabel('Amplitude')
legend('Original Signal','Reconstructed Signal')
grid on

%% Display Information
disp('-----');
disp('POLYPHASE DECIMATION & INTERPOLATION');
disp('-----');

fprintf('Original Signal Length = %d\n',length(x));
fprintf('Decimated Signal Length = %d\n',length(y_dec));
fprintf('Interpolated Signal Length = %d\n',length(y_int));

%% Separate Figure for Better Visualization
figure;

plot(t1,x,'b','LineWidth',2)
hold on
plot(t3,y_int,'r--','LineWidth',2)

```

```
title('Original vs Reconstructed Signal')
xlabel('Time (s)')
ylabel('Amplitude')
legend('Original Signal','Reconstructed Signal')
grid on
```

Output:



